SOLUTIONS: SYSTEMS IDENTIFICATION

1. Solution

a) We write:

$$p(t+1) = \frac{2}{3}\cdot p(t) - \frac{1}{3}\cdot p(t-1) + \eta(t)$$

$$z\cdot p(t) = \frac{2}{3}\cdot p(t) - \frac{1}{3}\cdot z^{-1}\cdot p(t) + \eta(t)$$

$$z\cdot p(t) - \frac{2}{3}\cdot p(t) + \frac{1}{3}\cdot z^{-1}\cdot p(t) = \eta(t)$$

$$\left(z - \frac{2}{3} + \frac{1}{3}\cdot z^{-1}\right)\cdot p(t) = \eta(t)$$

$$p(t) = \frac{1}{z - \frac{2}{3} + \frac{1}{3}\cdot z^{-1}}\cdot \eta(t)$$

$$p(t) = \frac{z}{z^2 - \frac{2}{3}\cdot z + \frac{1}{3}}\cdot \eta(t)$$

The process $p(\cdot)$ is stationary if and only if the roots of the polynomial $z^2 - \frac{2}{3}z + \frac{1}{3}$ are located strictly inside the unit circle:

$$z_{1,2} = \frac{\frac{2}{3} \pm \sqrt{\frac{4}{9} - 4\cdot 1\cdot \frac{1}{3}}}{2} = \frac{\frac{2}{3} \pm \sqrt{-\frac{8}{9}}}{2} = \frac{1}{3} \pm j\cdot \frac{\sqrt{2}}{3}$$

Since

$$|z_{1,2}| = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{\sqrt{2}}{3}\right)^2} = \frac{\sqrt{3}}{3} < 1$$

the process $p(\cdot)$ is stationary.

[3 Marks]

b) From the answer to Question 1-a), we get:

$$\mathbb{E}[p(t)] = \mathbb{E}[p(t+1)]$$

$$= \mathbb{E}\left[\frac{2}{3}\cdot p(t) - \frac{1}{3}\cdot p(t-1) + \eta(t)\right]$$

$$= \frac{2}{3}\cdot \mathbb{E}[p(t)] - \frac{1}{3}\cdot \mathbb{E}[p(t-1)] + \mathbb{E}[\eta(t)]$$

$$\mathbb{E}[p(t)]\cdot\left(1 - \frac{2}{3} + \frac{1}{3}\right) = 0$$

$$\mathbb{E}[p(t)] = 0$$

[3 Marks]

Answers

c) As $\mathbb{E}[p(t)] = 0$, we obtain:

$$\begin{aligned}
\gamma_p(0) &= \mathbb{E}\left[p(t)^2\right] \\
&= \mathbb{E}\left[\left(\frac{2}{3}\cdot p(t-1) - \frac{1}{3}\cdot p(t-2) + \eta(t-1)\right)^2\right] \\
&= \frac{4}{9}\cdot \underbrace{\mathbb{E}\left[p(t-1)^2\right]}_{\gamma_p(0)} + \frac{1}{9}\cdot \underbrace{\mathbb{E}\left[p(t-2)^2\right]}_{\gamma_p(0)} + \underbrace{\mathbb{E}\left[\eta(t-1)^2\right]}_{1/3} + \\
&\quad - \frac{4}{9}\cdot \underbrace{\mathbb{E}[p(t-1)\,p(t-2)]}_{\gamma_p(1)} + \frac{4}{3}\cdot \underbrace{\mathbb{E}[p(t-1)\cdot\eta(t-1)]}_{0} - \frac{2}{3}\cdot \underbrace{\mathbb{E}[p(t-2)\cdot\eta(t-1)]}_{0} \\
&= \frac{4}{9}\cdot\gamma_p(0) + \frac{1}{9}\cdot\gamma_p(0) + \frac{1}{3} - \frac{4}{9}\cdot\gamma_p(1)
\end{aligned}$$

Now:

$$\begin{aligned}
\gamma_p(1) &= \mathbb{E}[p(t)\cdot p(t-1)] \\
&= \mathbb{E}\left[\left(\frac{2}{3}\cdot p(t-1) - \frac{1}{3}\cdot p(t-2) + \eta(t-1)\right)\cdot p(t-1)\right] \\
&= \frac{2}{3}\cdot \underbrace{\mathbb{E}\left[p(t-1)^2\right]}_{\gamma_p(0)} - \frac{1}{3}\cdot \underbrace{\mathbb{E}[p(t-1)\cdot p(t-2)]}_{\gamma_p(1)} + \underbrace{\mathbb{E}[\eta(t-1)\cdot p(t-1)]}_{0} \\
&= \frac{2}{3}\cdot\gamma_p(0) - \frac{1}{3}\cdot\gamma_p(1)
\end{aligned}$$

Hence:

$$\begin{cases}
\gamma_p(0) = \frac{4}{9}\cdot\gamma_p(0) + \frac{1}{9}\cdot\gamma_p(0) + \frac{1}{3} - \frac{4}{9}\cdot\gamma_p(1) \\
\gamma_p(1) = \frac{2}{3}\cdot\gamma_p(0) - \frac{1}{3}\cdot\gamma_p(1)
\end{cases}$$

From the second equation, we obtain

$$\begin{aligned}
\gamma_p(1) &= \frac{2}{3}\cdot\gamma_p(0) - \frac{1}{3}\cdot\gamma_p(1) \\
\frac{4}{3}\cdot\gamma_p(1) &= \frac{2}{3}\cdot\gamma_p(0) \\
2\cdot\gamma_p(1) &= \gamma_p(0)
\end{aligned}$$

then

$$\begin{aligned}
\gamma_p(0) &= \frac{4}{9}\cdot\gamma_p(0) + \frac{1}{9}\cdot\gamma_p(0) + \frac{1}{3} - \frac{4}{9}\cdot\gamma_p(1) \\
2\cdot\gamma_p(1) &= \frac{4}{9}\cdot 2\cdot\gamma_p(1) + \frac{1}{9}\cdot 2\cdot\gamma_p(1) + \frac{1}{3} - \frac{4}{9}\cdot\gamma_p(1) \\
\gamma_p(1) &= \frac{1}{4}
\end{aligned}$$

and

$$\gamma_p(0) = 2\cdot\frac{1}{4} = \frac{1}{2}$$

[4 Marks]

Answers

d) From the answer to Question 1-a), we recall that

$$p(t) = \frac{z}{z^2 - \frac{2}{3}\cdot z + \frac{1}{3}} \cdot \eta(t)$$

Then:

$$\begin{aligned}
\Gamma_p(\omega) &= \left| \frac{e^{j\cdot\omega}}{e^{2\cdot j\cdot\omega} - \frac{2}{3}\cdot e^{j\cdot\omega} + \frac{1}{3}} \right|^2 \cdot \frac{1}{3} \\
&= \frac{\left|e^{j\cdot\omega}\right|^2}{\left| e^{2\cdot j\cdot\omega} - \frac{2}{3}\cdot e^{j\cdot\omega} + \frac{1}{3} \right|^2} \cdot \frac{1}{3} \\
&= \frac{1}{\left(e^{2\cdot j\cdot\omega} - \frac{2}{3}\cdot e^{j\cdot\omega} + \frac{1}{3} \right) \cdot \left(e^{-2\cdot j\cdot\omega} - \frac{2}{3}\cdot e^{-j\cdot\omega} + \frac{1}{3} \right)} \cdot \frac{1}{3} \\
&= \frac{1}{1 + \frac{4}{9} + \frac{1}{9} + \frac{1}{3}\cdot\left(e^{2\cdot j\cdot\omega} + e^{-2\cdot j\cdot\omega}\right) + \left(-\frac{2}{3}\cdot\frac{1}{3} - \frac{2}{3} \right)\cdot\left(e^{j\cdot\omega} + e^{-j\cdot\omega}\right)} \cdot \frac{1}{3} \\
&= \frac{1}{1 + \frac{4}{9} + \frac{1}{9} + \left(-\frac{2}{3}\cdot\frac{1}{3} - \frac{2}{3} \right)\cdot\left(e^{j\cdot\omega} + e^{-j\cdot\omega}\right) + \frac{1}{3}\cdot\left(e^{2\cdot j\cdot\omega} + e^{-2\cdot j\cdot\omega}\right)} \cdot \frac{1}{3} \\
&= \frac{1}{\frac{14}{9} - \frac{8}{9}\cdot 2\cdot\cos(\omega) + \frac{1}{3}\cdot 2\cdot\cos(2\cdot\omega)} \cdot \frac{1}{3} \\
&= \frac{1}{\frac{14}{9} - \frac{16}{9}\cdot\cos(\omega) + \frac{2}{3}\cdot\cos(2\cdot\omega)} \cdot \frac{1}{3} \\
&= \frac{3}{14 - 16\cdot\cos(\omega) + 6\cdot\cos(2\cdot\omega)}.
\end{aligned}$$

To provide a qualitative sketch of the spectrum, it is useful to compute some values of $\Gamma_p(\omega)$:

$$\Gamma_p(0) = \frac{3}{14 - 16\cdot\cos(0) + 6\cdot\cos(2\cdot 0)} = \frac{3}{4}$$

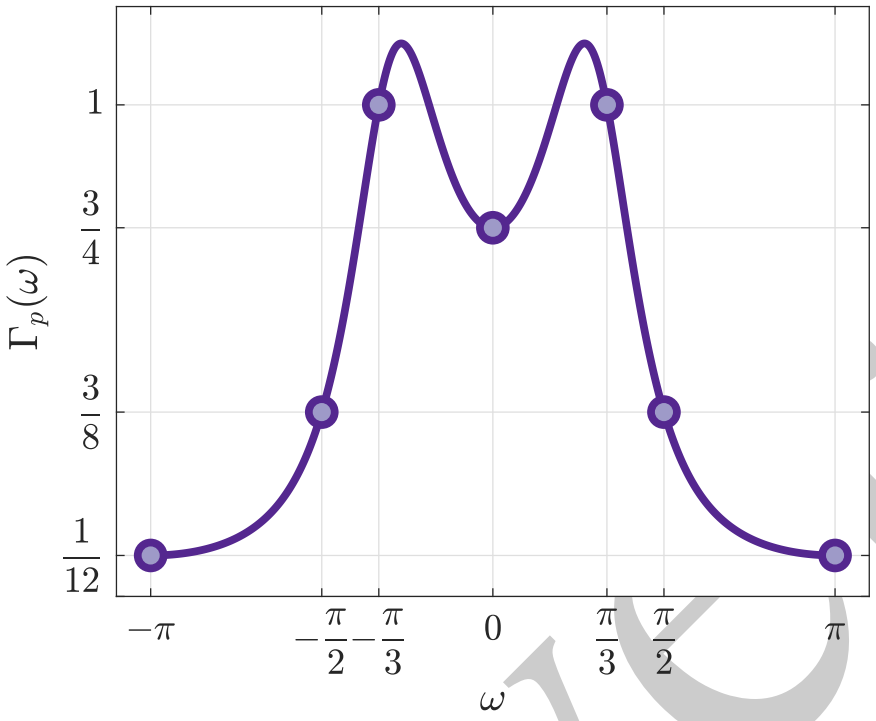
$$\Gamma_p\left(\frac{\pi}{3}\right) = \Gamma_p\left(-\frac{\pi}{3}\right) = \frac{3}{14 - 16\cdot\cos\left(\frac{\pi}{3}\right) + 6\cdot\cos\left(2\cdot\frac{\pi}{3}\right)} = 1$$

$$\Gamma_p\left(\frac{\pi}{2}\right) = \Gamma_p\left(-\frac{\pi}{2}\right) = \frac{3}{14 - 16\cdot\cos\left(\frac{\pi}{2}\right) + 6\cdot\cos\left(2\cdot\frac{\pi}{2}\right)} = \frac{3}{8}$$

$$\Gamma_p(\pi) = \Gamma_p(-\pi) = \frac{3}{14 - 16\cdot\cos\left(\frac{\pi}{2}\right) + 6\cdot\cos\left(2\cdot\frac{\pi}{2}\right)} = \frac{1}{12}$$

Answers

The diagram of $\Gamma_p(\omega)$ interpolating the values computed above is shown in the following figure.



[3 Marks]

e) The linear estimator has the form

$$\begin{aligned}\hat{p}(t) &= \alpha \cdot s_1(t) + \beta \cdot s_2(t) + \gamma \\ &= \alpha \cdot \left(\frac{5}{4} \cdot p(t) + \frac{1}{6} \cdot e_1(t)\right) + \beta \cdot \left(\frac{1}{2} \cdot p(t) + e_2(t)\right) + \gamma \\ &= p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta\right) + \frac{\alpha}{6} \cdot e_1(t) + \beta \cdot e_2(t) + \gamma\end{aligned}$$

and the estimator error is given by

$$\begin{aligned}\varepsilon(t) &= \hat{p}(t) - p(t) \\ &= p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta\right) + \frac{\alpha}{6} \cdot e_1(t) + \beta \cdot e_2(t) + \gamma - p(t) \\ &= p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1\right) + \frac{\alpha}{6} \cdot e_1(t) + \beta \cdot e_2(t) + \gamma\end{aligned}$$

The estimator is unbiased if $\mathbb{E}[\varepsilon(t)] = 0$. Hence, we compute

$$\begin{aligned}\mathbb{E}[\varepsilon(t)] &= \mathbb{E}\left[p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1\right) + \frac{\alpha}{6} \cdot e_1(t) + \beta \cdot e_2(t) + \gamma\right] \\ &= \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1\right) \cdot \underbrace{\mathbb{E}[p(t)]}_{0} + \frac{\alpha}{6} \cdot \underbrace{\mathbb{E}[e_1(t)]}_{1} + \beta \cdot \underbrace{\mathbb{E}[e_2(t)]}_{0} + \gamma \\ &= \frac{\alpha}{6} + \gamma\end{aligned}$$

Answers

Imposing $\mathbb{E}[\varepsilon(t)] = 0$, we obtain:

$$\gamma = -\frac{\alpha}{6}$$

Now, we can search the values of α and β such that the variance of the estimation error is minimised. Therefore, let us recall that the estimation error is given by

$$\begin{aligned}
\varepsilon(t) &= p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) + \frac{\alpha}{6} \cdot e_1(t) + \beta \cdot e_2(t) + \gamma \\
&= p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) + \frac{\alpha}{6} \cdot e_1(t) + \beta \cdot e_2(t) - \frac{\alpha}{6} \\
&= p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) + \frac{\alpha}{6} \cdot (e_1(t) - 1) + \beta \cdot e_2(t)
\end{aligned}$$

For simplicity, let us define

$$\begin{aligned}
v(t) &= e_1(t) - 1 \\
\mathbb{E}[v(t)] &= \mathbb{E}[e_1(t)] - 1 = 0 \\
\mathbb{E}\left[v(t)^2 \right] &= \mathbb{E}\left[(e_1(t) - 1)^2 \right] = 1
\end{aligned}$$

thus getting

$$\varepsilon(t) = p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) + \frac{\alpha}{6} \cdot v(t) + \beta \cdot e_2(t) \tag{1.1}$$

Since we imposed $\mathbb{E}[\varepsilon(t)] = 0$, the variance of the estimation error is given by

$$\begin{aligned}
\mathbb{E}\left[(\varepsilon(t))^2 \right] &= \mathbb{E}\left[\left(p(t) \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) + \frac{\alpha}{6} \cdot v(t) + \beta \cdot e_2(t) \right)^2 \right] \\
&= \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right)^2 \cdot \underbrace{\mathbb{E}\left[p(t)^2 \right]}_{1/2} + \frac{\alpha^2}{36} \cdot \underbrace{\mathbb{E}\left[v(t)^2 \right]}_{1} + \beta^2 \cdot \underbrace{\mathbb{E}\left[e_2(t)^2 \right]}_{1} + \\
&\quad + 2 \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) \cdot \frac{\alpha}{6} \cdot \underbrace{\mathbb{E}[p(t) \cdot v(t)]}_{0} + 2 \cdot \frac{\alpha}{6} \cdot \beta \cdot \underbrace{\mathbb{E}[v(t) \cdot e_2(t)]}_{0} + \\
&\quad + 2 \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) \cdot \beta \cdot \underbrace{\mathbb{E}[p(t) \cdot e_2(t)]}_{0} \\
&= \frac{1}{2} \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right)^2 + \frac{\alpha^2}{36} + \beta^2
\end{aligned}$$

The components of the gradient of $\mathbb{E}\left[(\varepsilon(t))^2 \right]$ are:

$$\begin{aligned}
\frac{\partial}{\partial \alpha} \mathbb{E}\left[(\varepsilon(t))^2 \right] &= \frac{\partial}{\partial \alpha} \left[\frac{1}{2} \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right)^2 + \frac{\alpha^2}{36} + \beta^2 \right] \\
&= \frac{1}{2} \cdot 2 \cdot \left(\frac{5}{4} \cdot \alpha + \frac{1}{2} \cdot \beta - 1 \right) \cdot \frac{5}{4} + \frac{2}{36} \cdot \alpha
\end{aligned}$$

Answers

$$
\begin{aligned}
&= \left(\frac{25}{16}+\frac{2}{36}\right)\cdot\alpha+\frac{5}{8}\cdot\beta-\frac{5}{4} \\
&= \frac{233}{144}\cdot\alpha+\frac{5}{8}\cdot\beta-\frac{5}{4} \\
\frac{\partial}{\partial\beta}\mathbb{E}\left[(\varepsilon(t))^2\right] &= \frac{\partial}{\partial\beta}\left[\frac{1}{2}\cdot\left(\frac{5}{4}\cdot\alpha+\frac{1}{2}\cdot\beta-1\right)^2+\frac{\alpha^2}{36}+\beta^2\right] \\
&= \frac{5}{8}\cdot\alpha+\left(\frac{1}{4}+2\right)\cdot\beta-\frac{1}{2} \\
&= \frac{5}{8}\cdot\alpha+\frac{9}{4}\cdot\beta-\frac{1}{2}
\end{aligned}
$$

Therefore, to find the minimum we have to solve the system of linear equations:

$$
\begin{cases}
\frac{233}{144}\cdot\alpha+\frac{5}{8}\cdot\beta-\frac{5}{4}=0 \\
\frac{5}{8}\cdot\alpha+\frac{9}{4}\cdot\beta-\frac{1}{2}=0
\end{cases}
$$

From the second equation, we have

$$
\alpha=\frac{4}{5}-\frac{18}{5}\cdot\beta \tag{1.2}
$$

then, the first equation becomes

$$
\begin{aligned}
\frac{233}{144}\cdot\left(\frac{4}{5}-\frac{18}{5}\cdot\beta\right)+\frac{5}{8}\cdot\beta-\frac{5}{4}&=0 \\
\left(\frac{5}{8}-\frac{233}{144}\cdot\frac{18}{5}\right)\cdot\beta+\frac{233}{144}\cdot\frac{4}{5}-\frac{5}{4}&=0 \\
-\frac{26}{5}\cdot\beta&=-\frac{2}{45} \\
\beta&=\frac{1}{117}
\end{aligned}
$$

and

$$
\alpha=\frac{4}{5}-\frac{18}{5}\cdot\beta=\frac{4}{5}-\frac{18}{5}\cdot\frac{1}{117}=\frac{10}{13} \tag{1.3}
$$

Finally, the minimum-variance unbiased estimator is:

$$
\begin{aligned}
\hat{p}(t) &= \alpha\cdot s_1(t)+\beta\cdot s_2(t)+\gamma \\
&= \frac{10}{13}\cdot s_1(t)+\frac{1}{117}\cdot s_2(t)-\frac{\alpha}{6} \\
&= \frac{10}{13}\cdot s_1(t)+\frac{1}{117}\cdot s_2(t)-\frac{5}{39}
\end{aligned}
$$

[7 Marks]

Answers

2. Solution

a) We immediately get:

$$\begin{cases} z\cdot x_1(t) = \dfrac{1}{10}\cdot x_1(t) + \dfrac{10}{3}\cdot u(t) + \dfrac{7}{2}\cdot v(t) \\ z\cdot x_2(t) = \dfrac{3}{5}\cdot x_1(t) + \dfrac{2}{5}\cdot x_2(t) + v(t) \\ y(t) = \dfrac{1}{2}\cdot x_2(t) \end{cases}$$

After some algebra, we obtain:

$$y(t) = \frac{1}{z^2 - \frac{1}{2}\cdot z + \frac{1}{25}}\cdot u(t) + \frac{1+\frac{1}{2}\cdot z}{z^2 - \frac{1}{2}\cdot z + \frac{1}{25}}\cdot v(t)$$

which is the requested ARMAX model.

[4 Marks]

b) Before computing the predictor, it is necessary to check if the process is written in canonical spectral form. To do so, we can focus on the stochastic part of the process

$$\widetilde{y}(t) = \frac{1+\frac{1}{2}\cdot z}{z^2 - \frac{1}{2}\cdot z + \frac{1}{25}}\cdot v(t)$$

Firstly, we notice that there is a zero equal to -2, and it is outside the unit circle. By applying an all-pass filter, we obtain:

$$\widetilde{y}(t) = \frac{\cancel{1+\frac{1}{2}\cdot z}}{z^2 - \frac{1}{2}\cdot z + \frac{1}{25}}\cdot\frac{1}{2}\cdot\frac{1+2\cdot z}{\cancel{1+\frac{1}{2}\cdot z}}\cdot v(t) = \frac{\frac{1}{2}+z}{z^2 - \frac{1}{2}\cdot z + \frac{1}{25}}\cdot v(t)$$

Furthermore, the numerator and denominator have to have the same degree:

$$\widetilde{y}(t) = \frac{\frac{1}{2}+z}{z^2 - \frac{1}{2}\cdot z + \frac{1}{25}}\cdot v(t) = \frac{z^{-1}+\frac{1}{2}\cdot z^{-2}}{1 - \frac{1}{2}\cdot z^{-1} + \frac{1}{25}\cdot z^{-2}}\cdot v(t) = \frac{1+\frac{1}{2}\cdot z^{-1}}{1 - \frac{1}{2}\cdot z^{-1} + \frac{1}{25}\cdot z^{-2}}\cdot z^{-1}\cdot v(t)$$

Since the numerator and the denominator are monic, to obtain the canonical spectral form of the process, we only need to redefine the white process as follows:

$$\begin{aligned} &e(t) = z^{-1}\cdot v(t) = v(t-1) \\ &e(\cdot) \sim WN(0,1) \end{aligned}$$

Therefore

$$\widetilde{y}(t) = \frac{1+\frac{1}{2}\cdot z^{-1}}{1 - \frac{1}{2}\cdot z^{-1} + \frac{1}{25}\cdot z^{-2}}\cdot e(t)$$

Answers

and the ARMAX process can be rewritten as

$$y(t) = \frac{z^{-2}}{1 - \frac{1}{2}\cdot z^{-1} + \frac{1}{25}\cdot z^{-2}} \cdot u(t) + \frac{1 + \frac{1}{2}\cdot z^{-1}}{1 - \frac{1}{2}\cdot z^{-1} + \frac{1}{25}\cdot z^{-2}} \cdot e(t)$$

with

$$A(z) = 1 - \frac{1}{2}\cdot z^{-1} + \frac{1}{25}\cdot z^{-2}$$
$$B(z) = z^{-2}$$
$$C(z) = 1 + \frac{1}{2}\cdot z^{-1}$$

Now, we carry out the long division to isolate the predictable part from the non-predictable one.

$$\begin{array}{lcccc|cccc} C(z) = & 1 & \frac{1}{2}\cdot z^{-1} & & & 1 & -\frac{1}{2}\cdot z^{-1} & \frac{1}{25}\cdot z^{-2} & = A(z) \\ \hline & -1 & +\frac{1}{2}\cdot z^{-1} & -\frac{1}{25}\cdot z^{-2} & & 1 & = Q_1(z) & & \\ \hline R_1(z) = & \backslash & z^{-1} & -\frac{1}{25}\cdot z^{-2} & & & & & \end{array}$$

Therefore the recursive equation of the one-step predictor is

$$C(z)\cdot\hat{y}(t|t-1) = R_1(z)\cdot y(t) + B(z)\cdot Q_1(z)\cdot u(t)$$
$$\left(1 + \frac{1}{2}\cdot z^{-1}\right)\cdot\hat{y}(t|t-1) = \left(z^{-1} - \frac{1}{25}z^{-2}\right)\cdot y(t) + z^{-2}\cdot 1\cdot u(t)$$
$$\hat{y}(t|t-1) + \frac{1}{2}\cdot\hat{y}(t-1|t-2) = y(t-1) - \frac{1}{25}y(t-2) + u(t-2)$$
$$\hat{y}(t|t-1) = y(t-1) - \frac{1}{25}y(t-2) - \frac{1}{2}\cdot\hat{y}(t-1|t-2) + u(t-2)$$

The prediction error is

$$\varepsilon_1(t) = Q_1(z)\cdot e(t) = e(t)$$

therefore the variance is

$$\mathrm{Var}[\varepsilon_1(t)] = \mathbb{E}\left[(\varepsilon_1(t) - \mathbb{E}[\varepsilon_1(t)])^2\right]$$

since the predictor is unbiased, we have

$$\mathrm{Var}[\varepsilon_1(t)] = \mathbb{E}\left[(e(t))^2\right] = 1$$

[8 Marks]

c) In order to obtain the second-order predictor, we need to carry out another step of the long division:

Answers

$$
\begin{array}{lccc|ccc|l}
C(z) = & 1 & \frac{1}{2}\cdot z^{-1} & & 1 & -\frac{1}{2}\cdot z^{-1} & \frac{1}{25}\cdot z^{-2} & = A(z) \\
\hline
 & -1 & +\frac{1}{2}\cdot z^{-1} & -\frac{1}{25}\cdot z^{-2} & 1 & z^{-1} & & = Q_2(z) \\
\hline
 & \backslash & z^{-1} & -\frac{1}{25}\cdot z^{-2} & & & & \\
\hline
 & & -z^{-1} & \frac{1}{2}\cdot z^{-2} & \frac{1}{25}\cdot z^{-3} & & & \\
\hline
R_2(z) = & & \backslash & \frac{23}{50}\cdot z^{-2} & \frac{1}{25}\cdot z^{-3} & & &
\end{array}
$$

Therefore, the recursive equation of the one-step predictor is given by

$$
\begin{aligned}
C(z)\cdot\hat{y}(t|t-1) &= R_2(z)\cdot y(t) + B(z)\cdot Q_2(z)\cdot u(t) \\
\left(1+\frac{1}{2}\cdot z^{-1}\right)\cdot\hat{y}(t|t-1) &= \left(\frac{23}{50}\cdot z^{-2} - \frac{1}{25}z^{-3}\right)\cdot y(t) + z^{-2}\cdot\left(1+z^{-1}\right)\cdot u(t) \\
\left(1+\frac{1}{2}\cdot z^{-1}\right)\cdot\hat{y}(t|t-1) &= \left(\frac{23}{50}\cdot z^{-2} - \frac{1}{25}z^{-3}\right)\cdot y(t) + z^{-2}\cdot\left(1+z^{-1}\right)\cdot u(t) \\
\hat{y}(t|t-1) + \frac{1}{2}\cdot\hat{y}(t-1|t-2) &= \frac{23}{50}\cdot y(t-2) - \frac{1}{25}\cdot y(t-3) + u(t-2) + u(t-3) \\
\hat{y}(t|t-1) &= \frac{23}{50}\cdot y(t-2) - \frac{1}{2}\cdot\hat{y}(t-1|t-2) - \frac{1}{25}\cdot y(t-3) + u(t-2) + u(t-3)
\end{aligned}
$$

The prediction error is

$$
\mathrm{Var}\left[\varepsilon_2(t)\right] = \mathbb{E}\left[\left(\varepsilon_2(t) - \mathbb{E}\left[\varepsilon_2(t)\right]\right)^2\right]
$$

and hence the variance is

$$
\mathrm{Var}\left[\varepsilon_2(t)\right] = \mathbb{E}\left[\left(\varepsilon_2(t) - \mathbb{E}\left[\varepsilon_2(t)\right]\right)^2\right]
$$

Since the predictor is unbiased, we finally get

$$
\mathrm{Var}\left[\varepsilon_2(t)\right] = \mathbb{E}\left[\left(e(t) + e(t-1)\right)^2\right] = 1 + 1 = 2
$$

[8 Marks]

Answers

3. Solution

a) First, we notice that the process $y(\cdot)$ is a stationary $MA(2)$ process. For this reason, the process has finite memory and

$$\gamma_y(\tau) = 0 \qquad \forall \tau > 2$$

Therefore, we only need to compute the values of $\gamma_y(-2)$, $\gamma_y(-1)$, $\gamma_y(0)$, $\gamma_y(1)$ and $\gamma_y(2)$. Recalling that the auto-correlation function is symmetric, we know that

$$\gamma_y(-2) = \gamma_y(2) \qquad \gamma_y(-1) = \gamma_y(1)$$

Then, we have

$$\begin{aligned}
\gamma_y(0) &= \mathbb{E}\left[(y(t))^2\right] \\
&= \mathbb{E}\left[\left(\frac{2}{3}\cdot\eta(t-2) + \frac{1}{3}\cdot\eta(t)\right)^2\right] \\
&= \frac{4}{9}\cdot\underbrace{\mathbb{E}\left[\eta(t-2)^2\right]}_{1/4} + \frac{4}{9}\cdot\underbrace{\mathbb{E}[\eta(t-3)\cdot\eta(t)]}_{0} + \frac{1}{9}\cdot\underbrace{\mathbb{E}\left[\eta(t)^2\right]}_{1/4} \\
&= \frac{4}{9}\cdot\frac{1}{4} + \frac{1}{9}\cdot\frac{1}{4} = \frac{5}{36} \\
\gamma_y(1) = \gamma_y(-1) &= \mathbb{E}[y(t)\cdot y(t-1)] \\
&= \mathbb{E}\left[\left(\frac{2}{3}\cdot\eta(t-2) + \frac{1}{3}\cdot\eta(t)\right)\cdot\left(\frac{2}{3}\cdot\eta(t-3) + \frac{1}{3}\cdot\eta(t-1)\right)\right] \\
&= \frac{4}{9}\cdot\underbrace{\mathbb{E}[\eta(t-2)\cdot\eta(t-3)]}_{0} + \frac{2}{9}\cdot\underbrace{\mathbb{E}[\eta(t)\cdot\eta(t-3)]}_{0} + \\
&\qquad + \frac{2}{9}\cdot\underbrace{\mathbb{E}[\eta(t)\cdot\eta(t-3)]}_{0} + \frac{1}{9}\cdot\underbrace{\mathbb{E}[\eta(t)\cdot\eta(t-1)]}_{0} \\
&= 0 \\
\gamma_y(2) = \gamma_y(-2) &= \mathbb{E}[y(t)\cdot y(t-2)] \\
&= \mathbb{E}\left[\left(\frac{2}{3}\cdot\eta(t-2) + \frac{1}{3}\cdot\eta(t)\right)\cdot\left(\frac{2}{3}\cdot\eta(t-4) + \frac{1}{3}\cdot\eta(t-2)\right)\right] \\
&= \frac{4}{9}\cdot\underbrace{\mathbb{E}[\eta(t-2)\cdot\eta(t-4)]}_{0} + \frac{2}{9}\cdot\underbrace{\mathbb{E}\left[\eta(t-2)^2\right]}_{1/4} + \\
&\qquad + \frac{2}{9}\cdot\underbrace{\mathbb{E}[\eta(t)\cdot\eta(t-4)]}_{0} + \frac{1}{9}\cdot\underbrace{\mathbb{E}[\eta(t)\cdot\eta(t-2)]}_{0} \\
&= \frac{2}{9}\cdot\frac{1}{4} = \frac{1}{18}
\end{aligned}$$

Therefore, the correlation function is given by

$$\gamma_y(\tau) = \begin{cases} \frac{5}{36} & |\tau| = 0 \\ 0 & |\tau| = 1 \\ \frac{1}{18} & |\tau| = 2 \\ 0 & |\tau| > 2 \end{cases}$$

Answers

b) According to the PEM criterion, we need to compute the 1-step predictor of the model as function of the parameters to be estimated. Since the model $\mathscr{M}_1$ is an $AR(2)$, the optimal predictor is

$$\hat{y}(t|t-1;\alpha) = \alpha \cdot y(t-2)$$

therefore, the prediction error is

$$\varepsilon_1(t;\alpha) = y(t) - \alpha \cdot y(t-2)$$

and the PEM cost function is

$$\begin{aligned}
J_1(\alpha) &= \mathbb{E}\left[(\varepsilon_1(t;\alpha))^2\right] \\
&= \mathbb{E}\left[(y(t) - \alpha \cdot y(t-2))^2\right] \\
&= \mathbb{E}\left[y(t)^2 - 2 \cdot \alpha \cdot y(t) \cdot y(t-2) + \alpha^2 \cdot y(t-2)^2\right] \\
&= \underbrace{\mathbb{E}\left[y(t)^2\right]}_{\gamma_y(0)} - 2 \cdot \alpha \cdot \underbrace{\mathbb{E}[y(t) \cdot y(t-2)]}_{\gamma_y(2)} + \alpha^2 \cdot \underbrace{\mathbb{E}\left[y(t-2)^2\right]}_{\gamma_y(0)} \\
&= \gamma_y(0) - 2 \cdot \alpha \cdot \gamma_y(2) + \alpha^2 \cdot \gamma_y(0) \\
&= \left(1 + \alpha^2\right) \cdot \gamma_y(0) - 2 \cdot \alpha \cdot \gamma_y(2)
\end{aligned}$$

where $\gamma_y(\cdot)$ is the auto-correlation function of the stochastic process $y(\cdot)$ that is used to generate the data. Plugging in the values computed previously, we obtain the cost function

$$J_1(\alpha) = \left(1 + \alpha^2\right) \cdot \frac{5}{36} - \alpha \cdot \frac{1}{9} = \alpha^2 \cdot \frac{5}{36} - \alpha \cdot \frac{1}{9} + \frac{5}{36}$$

In order to find the stationary point, we compute the derivative

$$\frac{d}{d\alpha} J_1(\alpha) = \frac{d}{d\alpha}\left(\alpha^2 \cdot \frac{5}{36} - \alpha \cdot \frac{1}{9} + \frac{5}{36}\right) = 2 \cdot \frac{5}{36} \cdot \alpha - \frac{1}{9} = \frac{5}{18} \cdot \alpha - \frac{1}{9}$$

and search for its roots

$$\begin{aligned}
\frac{d}{d\alpha} J_1(\hat{\alpha}) &= 0 \\
\frac{5}{18} \cdot \hat{\alpha} - \frac{1}{9} &= 0 \\
\hat{\alpha} &= \frac{2}{5}
\end{aligned}$$

Finally, it is possible to estimate λ^2 by evaluating the cost function using the calculated value of $\hat{\alpha}$:

$$\hat{\lambda}^2 = J_1(\hat{\alpha}) = \left(\frac{2}{5}\right)^2 \cdot \frac{5}{36} - \frac{2}{5} \cdot \frac{1}{9} + \frac{5}{36} = \frac{7}{60}$$

therefore, the estimated model is given by

$$y(t) = \frac{2}{5} \cdot y(t-2) + \xi(t) \qquad \xi(\cdot) \sim WN\left(0, \frac{7}{60}\right)$$

[6 Marks]

Answers

c) According to the PEM criterion, we need to compute the 1-step predictor of the model as function of the parameters to be estimated. Since the model $\mathscr{M}_2$ is an $AR(3)$, the optimal predictor is

$$\hat{y}(t|t-1;\delta) = \beta \cdot y(t-2) + \delta \cdot y(t-3)$$

therefore, the prediction error is

$$\varepsilon_2(t;\beta,\delta) = y(t) - \beta \cdot y(t-2) - \delta \cdot y(t-3)$$

and the PEM cost function is

$$\begin{aligned}
J_2(\beta,\delta) &= \mathbb{E}\left[\varepsilon_2(t;\beta,\delta)^2\right] \\
&= \mathbb{E}\left[(y(t) - \beta \cdot y(t-2) - \delta \cdot y(t-3))^2\right] \\
&= \underbrace{\mathbb{E}\left[y(t)^2\right]}_{\gamma_y(0)} + \beta^2 \cdot \underbrace{\mathbb{E}\left[y(t-2)^2\right]}_{\gamma_y(0)} + \delta^2 \cdot \underbrace{\mathbb{E}\left[y(t-3)^3\right]}_{\gamma_y(0)} + \\
&\quad - 2\beta \cdot \underbrace{\mathbb{E}[y(t)y(t-2)]}_{\gamma_y(2)} - 2\delta \cdot \underbrace{\mathbb{E}[y(t)y(t-3)]}_{\gamma_y(3)} + 2\beta\delta \cdot \underbrace{\mathbb{E}[y(t-2)y(t-3)]}_{\gamma_y(1)} \\
&= \gamma_y(0) + \beta^2 \cdot \gamma_y(0) + \delta^2 \cdot \gamma_y(0) - 2\beta \cdot \gamma_y(2) - 2\delta \cdot \gamma_y(3) + 2\beta\delta \cdot \gamma_y(1)
\end{aligned}$$

where $\gamma_y(\cdot)$ is the auto-correlation function of the stochastic process $y(\cdot)$ that is used to generate the data. Plugging in the values computed previously, we obtain the cost function

$$\begin{aligned}
J_2(\beta,\delta) &= \frac{5}{36} + \beta^2 \cdot \frac{5}{36} + \delta^2 \cdot \frac{5}{36} - 2\beta \cdot \frac{1}{18} - 2\delta \cdot 0 + 2\beta\delta \cdot 0 \\
&= \beta^2 \cdot \frac{5}{36} + \delta^2 \cdot \frac{5}{36} - \beta \cdot \frac{1}{9} + \frac{5}{36}
\end{aligned}$$

In order to find the stationary point, we can compute the gradient

$$\begin{aligned}
\frac{\partial}{\partial\beta} J_2(\beta,\delta) &= \frac{\partial}{\partial\beta}\left(\beta^2 \cdot \frac{5}{36} + \delta^2 \cdot \frac{5}{36} - \beta \cdot \frac{1}{9} + \frac{5}{36}\right) \\
&= 2 \cdot \frac{5}{36} \cdot \beta - \frac{1}{9} = \frac{5}{18} \cdot \beta - \frac{1}{9}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial\delta} J_2(\beta,\delta) &= \frac{\partial}{\partial\delta}\left(\beta^2 \cdot \frac{5}{36} + \delta^2 \cdot \frac{5}{36} - \beta \cdot \frac{1}{9} + \frac{5}{36}\right) \\
&= 2 \cdot \frac{5}{36} \cdot \delta = \frac{5}{18} \cdot \delta
\end{aligned}$$

and search for its roots

$$\begin{cases} \frac{5}{18} \cdot \hat{\beta} - \frac{1}{9} = 0 \\ \frac{5}{18} \cdot \hat{\delta} = 0 \end{cases} \Rightarrow \begin{cases} \hat{\beta} = \frac{2}{5} \\ \hat{\delta} = 0 \end{cases}$$

Finally, it is possible to estimate μ^2 by evaluating the cost function using the calculated values of $\hat{\beta}$ and $\hat{\delta}$:

$$\hat{\mu}^2 = J_2\left(\hat{\beta},\hat{\delta}\right) = \left(\frac{2}{5}\right)^2 \cdot \frac{5}{36} + 0^2 \cdot \frac{5}{36} - \frac{2}{5} \cdot \frac{1}{9} + \frac{5}{36} = \frac{7}{60}$$

Answers

therefore, the estimated model is

$$y(t) = \frac{2}{5} \cdot y(t-2) + e(t) \qquad\qquad e(\cdot) \sim WN\left(0, \frac{7}{60}\right)$$

[6 Marks]

d) According to the answers to Questions 3-b) and 3-c), it is worth noting that the two estimated models are the same, and hence they yield the same performance. More specifically, the performance can be evaluated computing the value of the cost function using the estimated parameters. In this particular case, these two values are the same ($7/60$). Therefore, the higher-order family of models $\mathscr{M}_2$ does not provide any additional advantage compared to the family of models $\mathscr{M}_1$. This is due to the fact that the data are generated using a process with $\gamma_y(1)$ and $\gamma_y(3)$ equal to 0. This means that the sample $y(t-3)$ is not correlated with $y(t)$ nor with samples that are correlated with $y(t)$. Therefore, it does not carry any useful information for the determination of $y(t)$.

[4 Marks]

Answers

4. Solution

a) Evaluating the model at time $k \cdot h$, we obtain

$$\frac{d}{dt}v(k\cdot h) = a\cdot\left(1-\left(\frac{v(k\cdot h)}{v_{ref}}\right)^4-\left(\frac{S}{d(k\cdot h)-p(k\cdot h)}\right)^2\right)$$

$$\frac{d}{dt}p(k\cdot h) = v(k\cdot h)$$

Then, using the Euler discretisation method, we get:

$$h^{-1}\cdot(v((k+1)\cdot h)-v(k\cdot h)) = a\cdot\left(1-\left(\frac{v(k\cdot h)}{v_{ref}}\right)^4-\left(\frac{S}{d(k\cdot h)-p(k\cdot h)}\right)^2\right)$$

$$h^{-1}\cdot(p((k+1)\cdot h)-p(k\cdot h)) = v(k\cdot h)$$

Since $h = 1s$, we write:

$$v((k+1)\cdot h) = v(k\cdot h)+a\cdot\left(1-\left(\frac{v(k\cdot h)}{v_{ref}}\right)^4-\left(\frac{S}{d(k\cdot h)-p(k\cdot h)}\right)^2\right)$$

$$p((k+1)\cdot h) = p(k\cdot h)+v(k\cdot h)$$

and we are able to simplify the notation as follows:

$$v(k\cdot h) = v(k)$$
$$p(k\cdot h) = p(k)$$
$$d(k\cdot h) = d(k)$$

thus obtaining

$$v(k+1) = v(k)+a\cdot\left(1-\left(\frac{v(k)}{v_{ref}}\right)^4-\left(\frac{S}{d(k)-p(k)}\right)^2\right) \tag{4.1}$$

$$p(k+1) = p(k)+v(k) \tag{4.2}$$

[4 Marks]

b) From Equation (4.2), we get

$$p(k+1)-p(k) = v(k) \tag{4.3}$$

Hence, Equation (4.1) becomes

$$v(k+1) = p(k+1)-p(k)+a\cdot\left(1-\left(\frac{p(k+1)-p(k)}{v_{ref}}\right)^4-\left(\frac{S}{d(k)-p(k)}\right)^2\right)$$

By shifting the equation by one time instant, we obtain

$$v(k) = p(k)-p(k-1)+a\cdot\left(1-\left(\frac{p(k)-p(k-1)}{v_{ref}}\right)^4-\left(\frac{S}{d(k-1)-p(k-1)}\right)^2\right)$$

Answers

Using Equation (4.3) a second time, we obtain

$$p(k+1)-p(k)=p(k)-p(k-1)+a\cdot\left(1-\left(\frac{p(k)-p(k-1)}{v_{ref}}\right)^4-\left(\frac{S}{d(k-1)-p(k-1)}\right)^2\right)$$

$$p(k+1)=2\cdot p(k)-p(k-1)+a\cdot\left(1-\left(\frac{p(k)-p(k-1)}{v_{ref}}\right)^4-\left(\frac{S}{d(k-1)-p(k-1)}\right)^2\right)$$

Finally, we can shift by one time instant again, to obtain

$$p(k)=2\cdot p(k-1)-p(k-2)+a\cdot\left(1-\left(\frac{p(k-1)-p(k-2)}{v_{ref}}\right)^4-\left(\frac{S}{d(k-2)-p(k-2)}\right)^2\right)$$

In order to obtain the requested form, it is useful to use some trivial algebra manipulation in order to obtain

$$\underbrace{p(k)-2\cdot p(k-1)+p(k-2)}_{\rho_0(k)}=\underbrace{a}_{\vartheta_1}\cdot\underbrace{1}_{\rho_1(k)}+\underbrace{\left(-\frac{a}{v_{ref}^4}\right)}_{\vartheta_2}\cdot\underbrace{(p(k-1)-p(k-2))^4}_{\rho_2(k)}+\underbrace{\left(-a\cdot S^2\right)}_{\vartheta_3}\cdot\underbrace{(d(k-2)-p(k-2))^{-2}}_{\rho_3(k)}$$

Therefore, the discretised model can be rewritten as

$$\rho_0(k)=\sum_{i=1}^{3}\vartheta_i\cdot\rho_i(k)$$

where

$$\begin{aligned}
&\rho_0(k)=p(k)-2\cdot p(k-1)+p(k-2)\\
&\rho_1(k)=1 && \vartheta_1=a\\
&\rho_2(k)=(p(k-1)-p(k-2))^4 && \vartheta_2=-\frac{a}{v_{ref}^4}\\
&\rho_3(k)=(d(k-2)-p(k-2))^{-2} && \vartheta_3=-a\cdot S^2
\end{aligned}$$

[8 Marks]

c) As shown in the previous answer, the discrete-time model can be rewritten as

$$\rho_0(k)=\sum_{i=1}^{3}\vartheta_i\cdot\rho_i(k)=\boldsymbol{\vartheta}^\top\cdot\boldsymbol{\rho}(k)$$

where

$$\boldsymbol{\rho}(k)=\begin{bmatrix}\rho_1(k)\\ \rho_2(k)\\ \rho_3(k)\end{bmatrix}\qquad\qquad\boldsymbol{\vartheta}=\begin{bmatrix}\vartheta_1\\ \vartheta_2\\ \vartheta_3\end{bmatrix}$$

The least-squares cost function that we need to minimize is

$$J(\boldsymbol{\vartheta})=\sum_{k=q}^{n}\left(\rho_0(k)-\boldsymbol{\vartheta}^\top\cdot\boldsymbol{\rho}(k)\right)^2$$

Answers

where n is the number of data at our disposal and q is the first usable. In order to minimize this cost function, we compute its gradient:

$$\begin{aligned}
\frac{\partial}{\partial \boldsymbol{\vartheta}} J(\boldsymbol{\vartheta}) &= \frac{\partial}{\partial \boldsymbol{\vartheta}} \sum_{k=q}^{m} \left(\rho_0(k) - \boldsymbol{\vartheta}^\top \cdot \boldsymbol{\rho}(k)\right)^2 \\
&= 2 \cdot \sum_{k=q}^{m} \left(\rho_0(k) - \boldsymbol{\vartheta}^\top \cdot \boldsymbol{\rho}(k)\right) \cdot \frac{\partial}{\partial \boldsymbol{\vartheta}} \left(\rho_0(k) - \boldsymbol{\vartheta}^\top \cdot \boldsymbol{\rho}(k)\right) \\
&= 2 \cdot \sum_{k=q}^{m} \left(\rho_0(k) - \boldsymbol{\vartheta}^\top \cdot \boldsymbol{\rho}(k)\right) \cdot \boldsymbol{\rho}(k)^\top \\
&= 2 \cdot \left(\sum_{k=q}^{m} \rho_0(k) \cdot \boldsymbol{\rho}(k)^\top - \sum_{k=q}^{m} \boldsymbol{\vartheta}^\top \cdot \boldsymbol{\rho}(k) \cdot \boldsymbol{\rho}(k)^\top\right) \\
&= 2 \cdot \left(\sum_{k=q}^{m} \rho_0(k) \cdot \boldsymbol{\rho}(k)^\top - \boldsymbol{\vartheta}^\top \cdot \sum_{k=q}^{m} \boldsymbol{\rho}(k) \cdot \boldsymbol{\rho}(k)^\top\right)
\end{aligned}$$

Now, we compute the estimate by equating to zero the gradient as follows:

$$\begin{aligned}
\frac{\partial}{\partial \boldsymbol{\vartheta}} J\left(\hat{\boldsymbol{\vartheta}}\right) &= 0 \\
2 \cdot \left(\sum_{k=q}^{m} \rho_0(k) \cdot \boldsymbol{\rho}(k)^\top - \hat{\boldsymbol{\vartheta}}^\top \cdot \sum_{k=q}^{m} \boldsymbol{\rho}(k) \cdot \boldsymbol{\rho}(k)^\top\right) &= 0 \\
\hat{\boldsymbol{\vartheta}}^\top \cdot \sum_{k=q}^{m} \boldsymbol{\rho}(k) \cdot \boldsymbol{\rho}(k)^\top &= \sum_{k=q}^{m} \rho_0(k) \cdot \boldsymbol{\rho}(k)^\top \\
\left[\sum_{k=q}^{m} \boldsymbol{\rho}(k) \cdot \boldsymbol{\rho}(k)^\top\right] \cdot \hat{\boldsymbol{\vartheta}} &= \left[\sum_{k=q}^{m} \rho_0(k) \cdot \boldsymbol{\rho}(k)\right] \\
\hat{\boldsymbol{\vartheta}} &= \left[\sum_{k=q}^{m} \boldsymbol{\rho}(k) \cdot \boldsymbol{\rho}(k)^\top\right]^{-1} \cdot \left[\sum_{k=q}^{m} \rho_0(k) \cdot \boldsymbol{\rho}(k)\right]
\end{aligned}$$

Therefore, we can find the value of $\boldsymbol{\vartheta}$ by solving the system of linear equations. Finally, we can compute the driver's parameters from the relationships

$$\vartheta_1 = a \qquad \vartheta_2 = -\frac{a}{v_{ref}^4} \qquad \vartheta_3 = -a \cdot S^2$$

that we recall from the answer to Question 4-b), thus getting:

$$a = \vartheta_1 \qquad v_{ref} = \left(-\frac{\vartheta_1}{\vartheta_2}\right)^{1/4} \qquad S = \sqrt{-\frac{\vartheta_3}{\vartheta_1}}$$

[8 Marks]

Answers